

VIEW

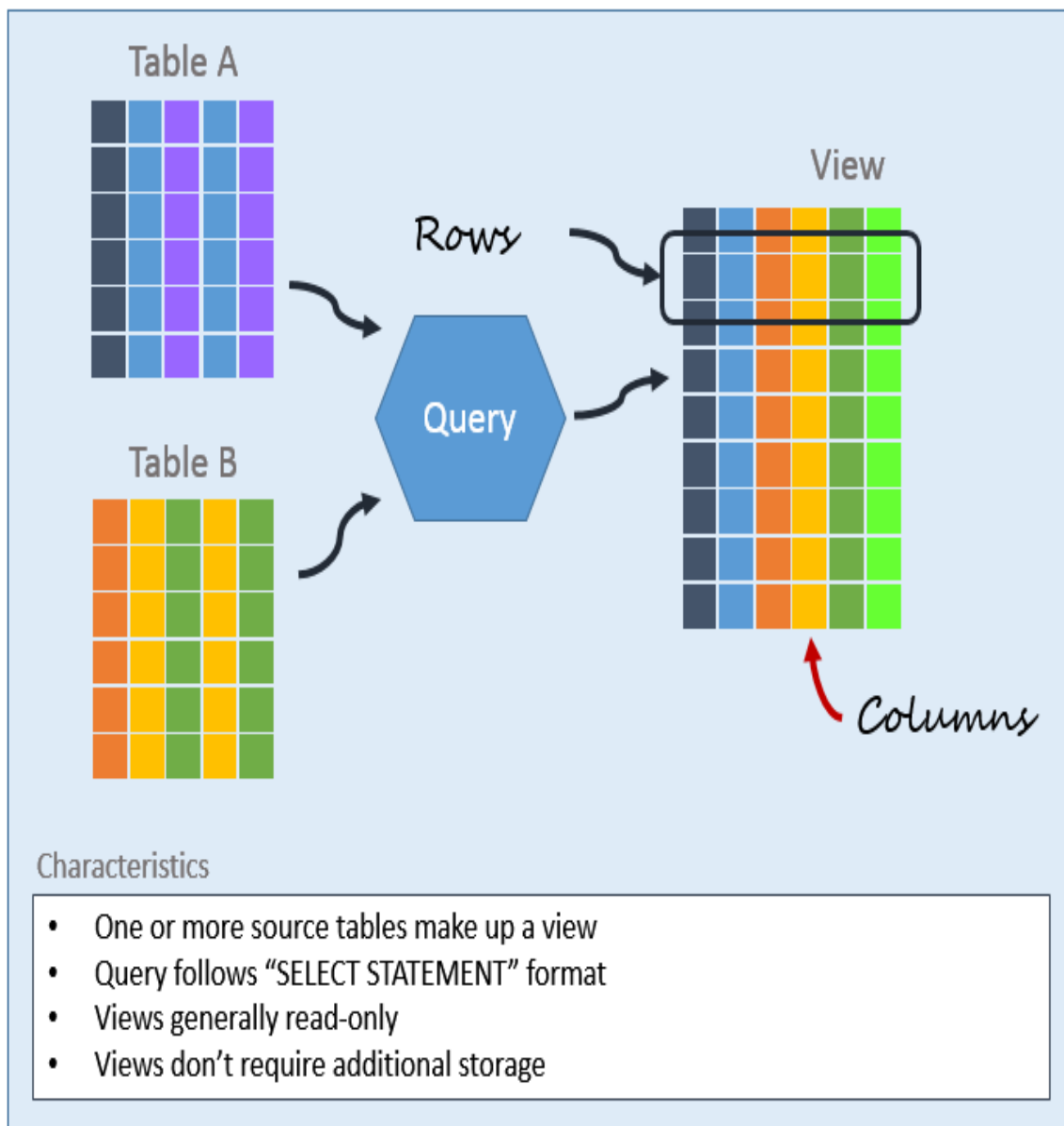
What is a View?

- A **View** in Oracle is a **virtual table** based on the result of a SQL query. It does not store data physically (except in materialized view). It stores only the **SELECT query definition**.
- When you query a view, Oracle executes the underlying query and displays the result.

Definition

A View is a logical representation of data from one or more tables.

Anatomy of a View



Benefits:

1. **Security** – Restrict access to specific columns/rows.
2. **Data Abstraction** – Hide complex joins and calculations.
3. **Simplification** – Make complex queries reusable.
4. **Consistency** – Provide standardized representation of data.
5. **Backward Compatibility** – Structure changes in base tables without affecting users.

Types of Views in Oracle

- **Simple View**
 - Based on a single table
 - No group functions
 - No GROUP BY
 - No DISTINCT
 - Usually updatable
- **Complex View**
 - Based on multiple tables
 - Contains:
 - JOIN
 - GROUP BY
 - Aggregate functions
 - DISTINCT
 - Generally not fully updatable
- **Inline View**
 - A subquery in FROM clause

Syntax of Creating View

```
CREATE [OR REPLACE] VIEW view_name
[(alias1, alias2, ...)]
AS
SELECT column1, column2, ...
FROM table_name
WHERE condition;
```

Example Table

```
CREATE TABLE EMPLOYEE (
  EMP_ID NUMBER PRIMARY KEY,
  EMP_NAME VARCHAR2(50),
  SALARY NUMBER(10,2),
  DEPT_ID NUMBER
);
```

Creating Views

A. Simple View Example

```
CREATE VIEW emp_view AS
SELECT emp_id, emp_name, salary
```

FROM employee;

Query the view

SELECT * FROM emp_view;

B. View with WHERE Clause

CREATE VIEW high_salary_emp AS

SELECT emp_id, emp_name, salary

FROM employee

WHERE salary > 50000;

C. View with Column Aliases

CREATE VIEW emp_details (id, name, income) AS

SELECT emp_id, emp_name, salary

FROM employee;

D. Complex View (Join Example)

CREATE VIEW emp_dept_view AS

SELECT e.emp_id, e.emp_name, d.dept_name

FROM employee e

JOIN department d

ON e.dept_id = d.dept_id;

E. View with GROUP BY

CREATE VIEW dept_salary AS

SELECT dept_id, SUM(salary) AS total_salary

FROM employee

GROUP BY dept_id;

Replacing a View

If view already exists:

CREATE OR REPLACE VIEW emp_view AS

SELECT emp_id, emp_name

FROM employee;

->Old definition is replaced.

Dropping a View

DROP VIEW view_name;

Example:

DROP VIEW emp_view;

Updating Data Through View

Syntax

UPDATE view_name

SET column_name = value

WHERE condition;

Example

```
UPDATE emp_view  
SET salary = 60000  
WHERE emp_id = 101;
```

Inserting Through View

```
INSERT INTO emp_view (emp_id, emp_name, salary)  
VALUES (105, 'Rahul', 45000);
```

Deleting Through View

```
DELETE FROM emp_view  
WHERE emp_id = 105;
```

When View is Updatable?

A view is updatable if:

- ✓ Based on single table
- ✓ No GROUP BY
- ✓ No DISTINCT
- ✓ No aggregate functions
- ✓ No JOIN
- ✓ No ROWNUM
- ✓ No subqueries in SELECT

WITH CHECK OPTION

Ensures inserted/updated rows satisfy view condition.

```
CREATE VIEW high_salary_emp AS  
SELECT emp_id, emp_name, salary  
FROM employee  
WHERE salary > 50000  
WITH CHECK OPTION;
```

If you try:

```
INSERT INTO high_salary_emp  
VALUES (106, 'Amit', 20000);
```

✗ ERROR — violates condition.

WITH READ ONLY

Prevents modification.

```
CREATE VIEW emp_read_only AS  
SELECT * FROM employee  
WITH READ ONLY;
```

Now:

```
UPDATE emp_read_only SET salary = 50000;
```

✗ ERROR

Data Dictionary for Views

To check views:

```
SELECT * FROM USER_VIEWS;  
SELECT * FROM ALL_VIEWS;
```

```
SELECT * FROM DBA_VIEWS;
```

Important Points to Take Care

⚠ **1. View does not store data (except materialized view)**

⚠ **2. Dropping base table:**

View becomes invalid.

⚠ **3. DML Restrictions:**

- Complex view may not allow update/insert/delete.

⚠ **4. Performance:**

- Views do not improve performance.
- Materialized views improve performance.

Difference: View vs Table

Feature	View	Table
Storage	No (logical)	Yes
Data	Derived	Actual
Security	High	Normal
DML	Limited	Full